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To cite this article: Anthony Chemero (2008) Self-Organization, Writ Large, Ecological Psychology, 20:3, 257-269, DOI: [10.1080/10407410802189372](https://doi.org/10.1080/10407410802189372)

To link to this article: <https://doi.org/10.1080/10407410802189372>



Published online: 09 Jul 2008.



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Self-Organization, Writ Large

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Imagine that nature is self-organizing all the way down: the processes and entities at any given scale are self-organized, autonomous systems; furthermore, the constituent parts of these self-organized, autonomous systems are themselves self-organized, autonomous systems. This is a view that people interested in self-organization and complexity, including ecological psychologists, would find congenial, and one can easily imagine that it encompasses scales of nature from chemical reactions to Gaia. In this essay I use non-well-founded set theory, also known as hyperset theory (Aczel, 1988; Barwise & Etchemendy, 1987; Barwise & Moss, 1996), to show how such a view of nature could be correct. I also show that this congenial view of nature as self-organized, autonomous systems at multiple scales has consequences for our understanding of affordances.

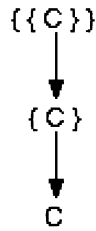
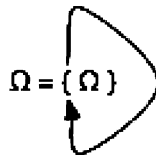
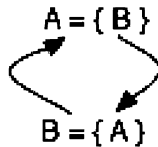
HYPERSETS, COMPLEXITY, AND AUTONOMY

Hyperset theory is a form of set theory that allows sets to have themselves as members, whether directly or indirectly. For example, the sets $\Omega = \{\Omega\}$ is illegal in standard set theory but legal in hyperset theory. The same is true for the circularly defined sets $A = \{B\}$ and $B = \{A\}$, in which A and B are indirectly members of themselves. The difference between sets and hypersets can be put in

*This is the third paper in the series, "Philosophical Issues in Self-organization as a Framework for Ecological Psychology," based on presentations given at the University of Connecticut, September 20–21, 2007.

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terms of graph theory. One draws graphs of sets by drawing an arrow from each set to all of its members. Thus sets $\{C\}$ and the $\{\{C\}\}$ have graphs as depicted in Figures 1 and 2, respectively; hypersets Ω and $\{A, B\}$ are graphed in Figures 3 and 4. In terms of graphs, we can put the difference between sets and hypersets as follows: sets have graphs without cycles; hypersets have graphs. Thus every set is a hyperset but not vice versa. This additional richness, the inclusion of sets with circular graphs, makes hyperset theory a more powerful modeling tool than standard set theory. Whereas set theory can model simplified, artificial languages and systems, hyperset theory can model complex, natural languages (Barwise &

FIGURE 1 Graphs of the sets $\{C\}$ and the $\{\{C\}\}$.FIGURE 2 Graphs of the sets $\{C\}$ and the $\{\{C\}\}$.FIGURE 3 Graphs of the hyperset Ω and the hyperset system A and B .FIGURE 4 Graphs of the hyperset Ω and the hyperset system A and B .

Etchemendy, 1987) and complex, natural systems (Chemero & Turvey, 2007a, 2008).

Of particular significance are demonstrations (a) that natural systems whose hyperset graphs contain loops are *complex* in the Rosen (1991, 2000) sense of complexity and (b) that hyperset graphs can be used to determine whether systems are *autonomous* (Chemero & Turvey, 2007a). Autonomous systems are systems in which every function or process in a system is the output of some other system or process. Chemero and Turvey (2008) modeled systems so that their efficient causal processes are functions. If each function in the model is the output of some other function in the model, then each of the system's efficient causes is also a product of the system. If this is the case, the system is what Rosen (1991, 2000) calls *closed to efficient cause*, that is, it produces the processes that maintain it. We proposed that this property just is autonomy. Thus, when every function in a hyperset graph of a system is the output of some other function, the graphed system is autonomous.

We can see how this works by looking at something like a definition of self-organized, autonomous systems provided by Kelso and Engström (2006): "Self-organizing dynamics creates and constrains functionally meaningful information and functionally meaningful information constrains self-organizing dynamics" (p. 198). We can render this relation between self-organizing dynamics (SOD) and functionally meaningful information (FMI) in terms of functional mappings. Self-organizing dynamics among the parts of a system create functionally meaningful information, so for a system of just two components, SOD is a function that uses the parts of a system as input and has FMI as output.

$$\text{SOD (Part 1, Part 2)} = \text{FMI}$$

Similarly, FMI is a function that takes the parts as input and has SOD as an output.

$$\text{FMI (Part 1, Part 2)} = \text{SOD}$$

We can render these functional relations as ordered pairs, with the input as the first member of the pair and the output as the second member of the pair.

$$\text{SOD} = \langle \{\text{Part 1, Part 2}\}, \text{FMI} \rangle \text{ and}$$

$$\text{FMI} = \langle \{\text{Part 1, Part 2}\}, \text{SOD} \rangle.$$

Figure 5 is a hyperset graph of these ordered pairs. Note that a standard convention is followed in that ordered pairs are represented as Kuratowska pairs, that is, as a singleton set of the first member of the ordered pair and as a set that contains both members of the pair. For example, $\langle E, F \rangle$ would be rendered as $\{E\}$ and $\{E, F\}$. Looking at this graph, we can see first of all that the graph contains

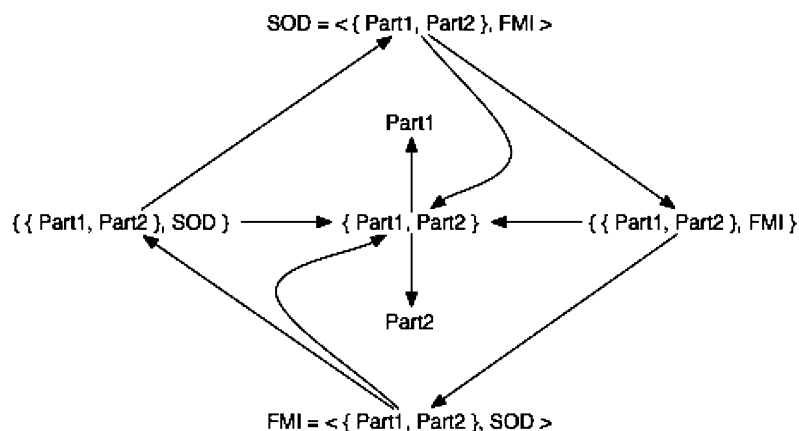


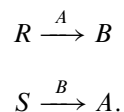
FIGURE 5 Graph of the relationship between self-organizing dynamics (SOD) and functionally meaningful information (FMI).

cycles. So the system graphed, the relation between SOD and FMI, is a complex system. Furthermore, notice that the system's two functions, SOD and FMI, are the outputs of one another. This indicates that the system is autonomous.

The view of nature as self-organized, autonomous systems at multiple scales implies that hyperset graphs of systems at every scale should have the same structure as Figure 5. In the next section, I present models of entities at other scales: we look at hyperset graphs of chemical reactions, cells, organisms, and animal-environment systems. Given these systems and their hyperset graphs are discussed in detail elsewhere (Chemero & Turvey 2007a, 2007b, and 2008), I discuss them only briefly here.

REALLY SMALL: COLLECTIVELY AUTOCATALYTIC CHEMICAL REACTIONS

Kauffman (1993) discusses collectively autocatalytic sets of chemical reactions as necessary precursors to living systems and argues that nature is self-organizing at the scale of chemical reactions. A collectively autocatalytic system of reactions is a set of reactions in which every reaction has a catalyst that is also the product of some other reaction in the network. Consider these two reactions:



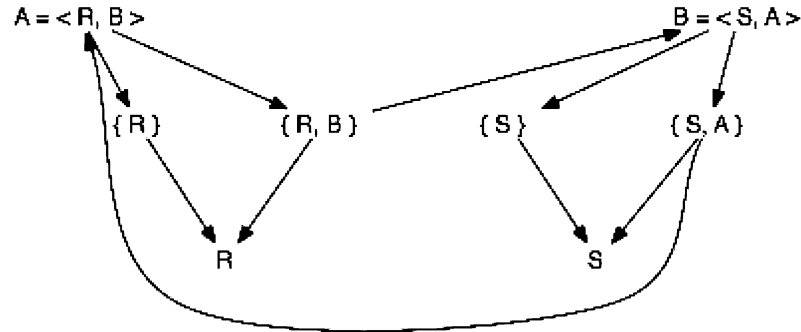


FIGURE 6 Graph of simplified autocatalytic network of reactions.

Because A catalyzes the production of B and vice versa, this is a collectively autocatalytic set of reactions. Figure 6 is a hyperset graph of this set of reactions. The graph was drawn by taking each of the catalysts A and B to be causes in the system and hence modelable as functions from raw materials to their products. So $A = \langle R, B \rangle$ and $B = \langle S, A \rangle$. The graph has a loop, so the collectively autocatalytic system is a complex system. Because both catalysts are the outputs of other functions, the collectively autocatalytic set of reactions is autonomous.

SLIGHTLY LARGER: CELLS

Maturana and Varela (1980) argue that cells are the simplest case of what they call autopoietic systems. *Autopoietic* literally means “self-producing.” Autopoietic systems are systems in which (a) the effects of the system are confined to the system and (b) the effects of a system *constitute* the system. In a cell, for example, metabolic processes create a cell membrane and the cell membrane bounds, and hence makes possible, the metabolic processes. Figure 7 is a hyperset graph of a simplified cell. In it, the metabolism is taken to be a causal process taking raw material as input and producing the membrane as output. So

$$\text{metabolism} = \langle \text{materials}, \text{membrane} \rangle.$$

The membrane is taken to be a causal process that uses raw materials to maintain the metabolism. So

$$\text{membrane} = \langle \text{materials}, \text{metabolism} \rangle.$$

Notice that the graph has a loop, so the system depicted is complex. Furthermore, because both functions depicted in the graph are the outputs of other

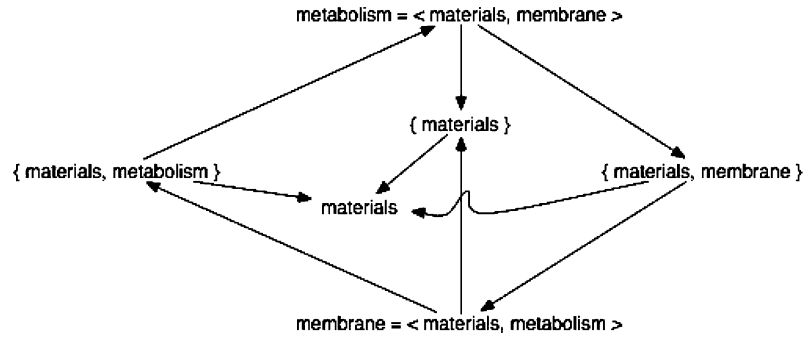


FIGURE 7 Graph of simplified cell.

functions, hence at the tail of arrows, the system is autonomous. So, one might argue, cells are self-organized, autonomous systems composed of self-organized systems at a smaller scale.

MESOSCOPIC: ORGANISMS

Cells are the building blocks of living things. Are living things also self-organized, autonomous systems? Rosen (1991) argues that the simplest living things must be *metabolism-repair systems*. A metabolism-repair system is a collection of three interconnected functions. First, there is Metabolism, which takes raw materials as input and produces Behavior as output. Behavior is also a function that takes Metabolism as an input and produces a Repair function. The Repair function uses the organism's Behavior as input to maintain the system's Metabolism. These functions are as follows:

$$\text{Metabolism} = \langle \text{Materials}, \text{Behavior} \rangle$$

$$\text{Behavior} = \langle \text{Metabolism}, \text{Repair} \rangle$$

$$\text{Repair} = \langle \text{Behavior}, \text{Metabolism} \rangle$$

The overall idea is that Metabolism produces the Behavior possible, part of the organism's Behavior is the creation of a Repair function, and the Repair function keeps the Metabolism going. Rosen (1991) argued that this three-function system is *closed to efficient cause* in that every function (efficient cause) in the system is produced by the system. Chemero and Turvey (2008) argued that closure to efficient cause is autonomy. The hyperset graph of the metabolism-repair system

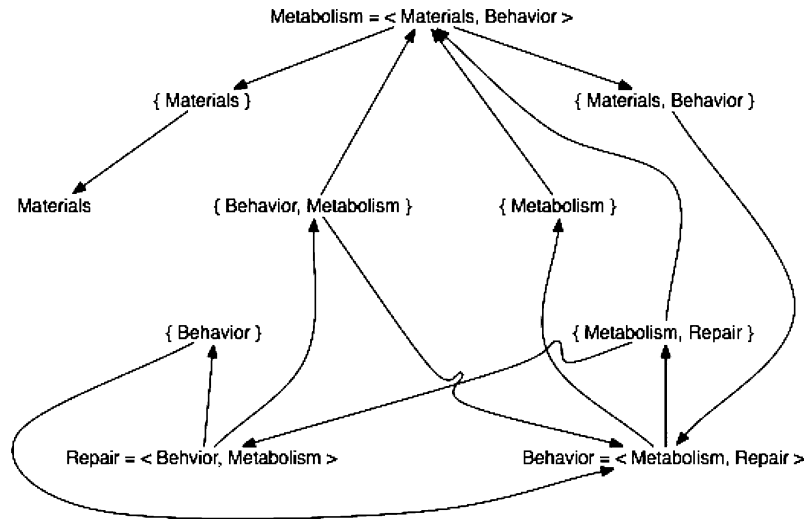


FIGURE 8 Graph of metabolism-repair system.

(Figure 8) bears this out. We can see from the graph that there are loops so the system is complex. We can also see that every function is the output of some other function, so the system is autonomous.

EXTENDED: ANIMAL-ENVIRONMENT SYSTEMS

Ecological psychologists have long been eager to include psychology among the sciences of self-organization and complexity. As part of this project, they have been instrumental in bringing nonlinear dynamics to the study of behavior (Kugler, Kelso, & Turvey, 1980; Warren, 2006). They also argue that the unit of study for psychology should be unified animal-environment systems, especially understood in terms of affordances (Gibson, 1979). Since Gibson introduced the notion, there have been two main approaches to understanding affordances. Either they are taken to be dispositional properties of the environment and are complemented by dispositional properties of animals (Turvey, 1992) or they are taken to be higher-order or relational properties of animal-environment systems (Chemero, 2003; Stoffregen, 2003). For present purposes, the disposition and relational views are not different from one another. (Indeed, Chemero & Turvey, 2007a, argue that the views are nearly identical.) Given this, I present my own formalization of affordance. The same conclusions follow from other relational views and from the dispositional view.

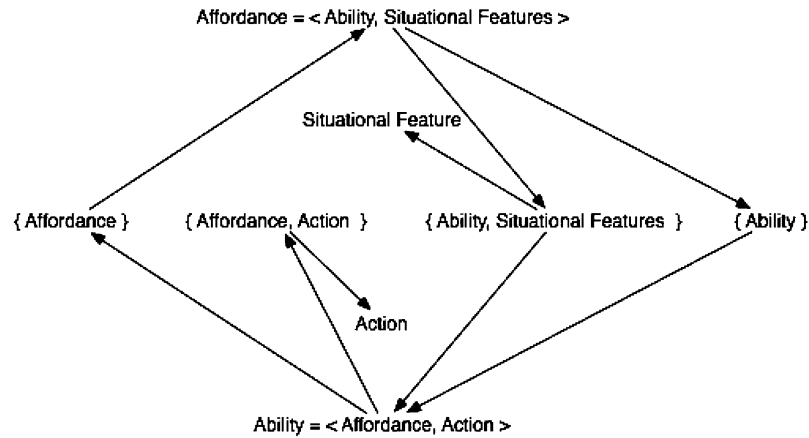


FIGURE 9 Graph of affordances as defined in Chemero 2003.

In Chemero (2003), I define *affordances* as relations between features of the environmental situations and the abilities of animals. So an affordance is a function or mapping from abilities (e.g., the ability to read) and situational features (e.g., the presence of text, appropriate lighting conditions, and so on). Abilities are also relations; they are mappings from affordances (e.g., legibility) and actions (e.g., reading). So to formalize the notion *affordance*, we need two functions.

$$\text{Affordance} = \langle \text{Ability}, \text{Situational Features} \rangle$$

$$\text{Ability} = \langle \text{Affordance}, \text{Action} \rangle$$

Figure 9 is the hyperset graph of this system of functions. Notice that because the graph contains loops, this graph depicts a complex system. Notice too that none of the system's functions are the outputs of other functions. Thus the graph does *not* depict a self-organizing, autonomous system. Indeed, none of the extant views of affordances of which I am aware make it the case that animal-environment systems are self-organizing, autonomous systems. (See the hyperset graphs in Chemero & Turvey, 2007a, 2007b, for confirmation.)

THE DILEMMA

This failure of the affordances of animal-environment systems to be self-organizing, autonomous systems leaves ecological psychologists in a dilemma. One option is to acknowledge that animal-environment systems are not self-organ-

izing, autonomous systems. As a consequence of choosing this option, ecological psychologists must give up the congenial view of nature as self-organizing autonomous systems at multiple scales and acknowledge that animals and their environments are not as tightly bound as we have, since Gibson, believed. Furthermore, ecological psychologists must acknowledge that psychology should not be allied with the sciences of complexity. I take it this option as akin to falling on one's sword, and wholly unacceptable. The other option is to admit that all extant conceptions of affordances—the central notion of ecological psychology—are inadequate. In other words, we ecological psychologists have failed to understand the central notion of ecological psychology.

I am quite certain that the latter of these options is correct—we do not understand affordances. Although far less than certain, I am also fairly confident about the problem with extant conceptions of affordances. We can see the problem with extant conceptions of affordance by comparing them with the self-organized, autonomous entities that we have looked at on smaller scales. The key difference is that collective autocatalysis, autopoiesis, and metabolism-repair are all processes, and the theoretical models of them are dynamic. In contrast, whether thought of as dispositions or as relations, the current conceptions of affordances take them to be static and model them in terms of how they are constituted. We need a conception of affordances according to which the affordances of the animal-environment system are dynamic relationships between animals and their environments. We need Affordances, Version 2.0.

AFFORDANCES 2.0

I don't yet have a full theory of Affordances 2.0, and certainly not enough of a formalization to produce hyperset graphs, but the following is a beginning. Step one is considering the interaction over time between an animal's sensorimotor abilities, that is, its embodied capacities for perception and action,¹ and its niche, that is, the set of affordances available to it. This is depicted in Figure 10, which is not a hyperset graph. Over developmental time, an animal's sensorimotor abilities select its niche—the animal will become selectively sensitive to information relevant to the things it is able to do. Also over developmental time, the niche will strongly influence the development of the animal's ability to perceive and act. Over the shorter time scales of behavior, the animal's sensorimotor abilities manifest themselves in embodied action that causes changes in the layout of

¹The phrase "sensorimotor abilities" is borrowed from enactivist cognitive science and is used here in order to illustrate the strong affinities between enactivism and ecological psychology. The term is *not* intended to imply that perception is mediated by sensation or that action is mediated by motor commands. Affordances 2.0 are not representationalist.

available affordances, and these affordances will change the way abilities are exercised in action. The key point here is that affordances and abilities are not just defined in terms of one another as in the dispositional and relational views discussed earlier but causally interact in real time and are causally dependent on one another.

There are four presently discernible consequences of this initial attempt to reconceptualize affordances. First, this is not so much a new way of understanding affordances as a critique of prior attempts to come up with a *definition* of the term *affordance*. Ecological psychologists have always been aware of, indeed keenly interested in, the interaction of affordances and abilities in real time. As noted previously, many ecological psychologists study perception and action dynamically. Affordances 2.0 is an attempt to develop a theoretical understanding of affordances that is more in line with the experimental and explanatory practices of ecological psychologists.

Second, notice that this reconceptualization of affordances just is a variety of niche construction that occurs over shorter time scales and in which the constructed niche is an animal's individual behavioral, cognitive, and phenomenological niche. In more standard biological niche construction, the activity of some population of organisms alters, sometimes dramatically, the population's own ecological niche as well as those of other organisms (Odling-Smee, Laland, & Feldman, 2003). These animal-caused alterations to niches have profound and wide-reaching effects over evolutionary time. And, indeed, the population of organisms and the niche are so tightly coupled that Griffiths and Gray (2001) recommend that they form a unified developmental system that is to be modeled with just one variable $\mathbb{E}$, whose dynamics is specified in the following equation,

$$d\mathbb{E}_{\text{pop}}/dt = f(\mathbb{E}_{\text{pop}}, E),$$

in which $\mathbb{E}_{\text{pop}}$ is the coupled organism-niche system for the population and E is the physical environment. The variety of niche construction sketched in Affordances 2.0 proposes an equally tightly coupled system animal-environment system. However, it differs from the much-discussed biological case in two ways. First, the constructed niche is for an individual organism, not for a population. Second, it occurs over shorter time scales—an animal's activities alter the world as the animal experiences it, and these alterations to the phenomenological-cognitive-behavioral niche, in turn, affect the animal's behavior and development of its abilities to perceive and act, which further alters the phenomenological-cognitive-behavioral niche, and on and on. Affordances 2.0, therefore, makes clear the connections between ecological psychology and its natural allies in biology, that is, proponents of developmental systems and niche construction.

Third, this reconceptualization of affordances is explicitly formulated to make the natural but largely unmade connections between ecological psychology and

another natural ally: the burgeoning enactivist movement in the cognitive sciences (Thompson, 2007; Varela, Thompson, & Rosch 1991). Figure 11 is an expanded version of Figure 10, expanded to show the connection between organisms and sensorimotor coupling, as understood by enactivists, and Affordances 2.0. (Figure 11 is also not a hyperset graph.) Enactivists view the organism as a self-organizing, autonomous, autopoietic system. In this system, the nervous system generates neuronal assemblies that make sensorimotor abilities possible and these sensorimotor abilities modulate the dynamics of the nervous system. Combining Affordances 2.0 with enactivist studies of the organism makes for a fully dynamical science of the entire brain-body-environment system: non-representational neurodynamic studies of the nervous system and sensorimotor abilities (Cosmelli, Lachaux, & Thompson, 2007; Thompson & Varela, 2001) match up with ecological psychological studies of affordances and sensorimotor abilities. (See Chemero & Silberstein, 2008; Chemero, in press, for more on the connections between enaction and ecological psychology.)

Fourth, and most important for current purposes, the dynamical science of the animal-environment system shown in Figure 11 shows how it might be the case that animal-environment systems are self-organized, autonomous systems. Much work remains to be done to show that this is indeed the case. First, Affordances 2.0 must be worked out in sufficient detail so that it can be shown that affordances and abilities constitute a self-organizing, autonomous system.

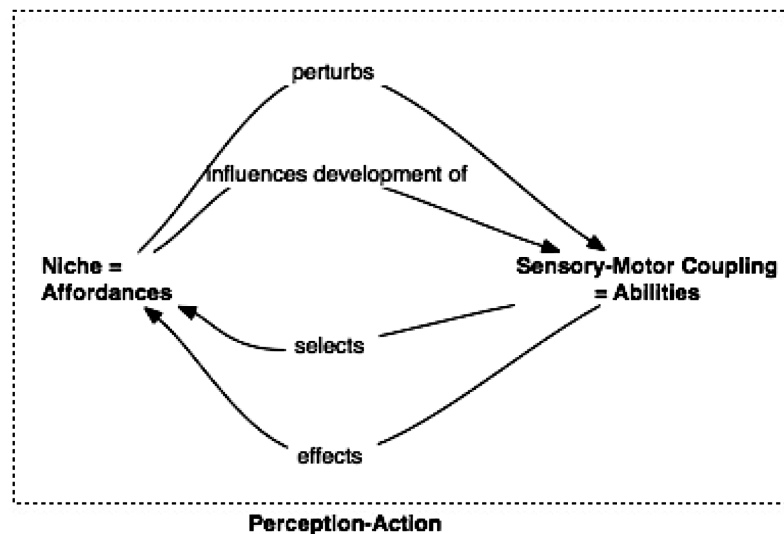


FIGURE 10 Depiction of dynamic relationships between affordances and abilities in perception-action.

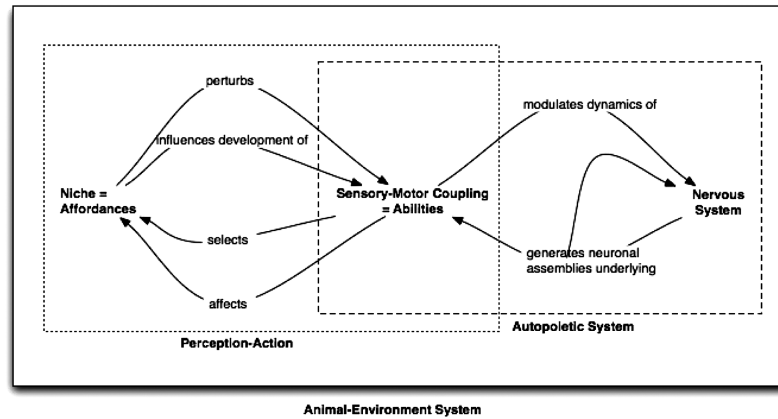


FIGURE 11 Depiction of dynamic relationships among affordances, abilities and the nervous system in an animal-environment system.

If that turns out to be so, then it must be shown that the affordance-ability self-organizing, autonomous system and the autopoietic nervous system jointly constitute a higher order self-organizing, autonomous system.

ACKNOWLEDGMENTS

Thanks to Michael Silberstein, whose comments on an earlier draft steered me away from disastrous oversimplification, and to Mike Turvey, whose comments helped me to be clearer.

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